Group 4 Project Specification Doc
RatCache

Title: RatCache

Overview:
Ratbin is a lightweight, web-based pastebin that lets users quickly share plain text, such as code snippets, error logs, configuration files, and notes, through a short, shareable link. Users can paste anonymously or create an account to manage their pastes, control who can see them, and set when they expire.

Scope:
Public general/dev pastebin with a web interface as well as CLI interface. Optional account creation, and server side encryption for users with accounts.

Functional Requirements

Priority follows MoSCoW: Must is required for v1.0, Should is planned if time allows, and Could is a stretch goal.

3.1 User Authentication and Account Creation

ID	Requirement	Priority
FR-1.1	The system shall let a visitor register with a username, email, and password.	Must
FR-1.2	The system shall reject a username or email that is already registered.	Must
FR-1.3	The system shall let a registered user sign in and sign out.	Must
FR-1.4	The system shall keep a signed-in user signed in through a session token until they sign out or the token expires.	Must
FR-1.5	The system shall let a signed-in user change their password.	Should
FR-1.6	The system shall let a user delete their account, and all of their pastes with it.	Should
FR-1.7	The system shall let a user reset a forgotten password through a link sent to their email.	Could
FR-1.8	The system shall let a signed-in user create and revoke API tokens for the CLI.	Must

3.2 Paste CRUD

ID	Requirement	Priority
FR-2.1	The system shall let any user create a paste of plain text or code, with an optional title.	Must
FR-2.2	The system shall give each new paste a unique, random, URL-safe ID and return its link.	Must
FR-2.3	The system shall show a paste to anyone who opens its link, as long as its visibility allows it and it has not expired.	Must
FR-2.4	The system shall offer a raw view of each paste that returns only the text, as text/plain.	Must
FR-2.5	The system shall let a registered user edit the content, title, language, visibility, and expiration of their own pastes.	Must
FR-2.6	The system shall let a registered user delete their own pastes.	Must
FR-2.7	The system shall list a registered user's pastes, newest first.	Must
FR-2.8	The system shall give the creator of an anonymous paste a one-time delete link.	Could
FR-2.9	The system shall let a user copy a paste's link or content with one click.	Should

3.3 Visibility and Expiration

ID	Requirement	Priority
FR-3.1	The system shall support three visibility levels: public (listed on the recent-pastes page), unlisted (link only), and private (owner only).	Must
FR-3.2	The system shall allow private pastes only for registered users.	Must
FR-3.3	The system shall let a user set an expiration of 10 minutes, 1 hour, 1 day, 1 week, 1 month, or never.	Must
FR-3.4	The system shall offer a burn-after-reading option that deletes a paste after its first view.	Could
FR-3.5	The system shall stop serving an expired paste and permanently delete it within 1 hour of its expiration time.	Must
FR-3.6	The system shall show a page of the 20 most recent public pastes.	Should

3.4 Syntax Highlighting

ID	Requirement	Priority
FR-4.1	The system shall let a user choose a paste's language from a list, or choose plain text.	Must
FR-4.2	The system shall syntax-highlight a paste in the web view according to its language.	Must

FR-4.3	The system shall support at least: C, C++, Python, JavaScript, Java, Go, Rust, Bash, SQL, HTML, CSS, JSON, YAML, and Markdown.	Must
FR-4.4	The system shall show line numbers in the web view.	Should

3.5 Web Portal

ID	Requirement	Priority
FR-5.1	The system shall provide a home page where a user can paste text and set its title, language, visibility, and expiration in one form.	Must
FR-5.2	The system shall provide a paste view page, a "my pastes" dashboard, sign-in and registration pages, and an account settings page.	Must
FR-5.3	The system shall work at screen widths from 360 px (phone) to 1920 px (desktop).	Should
FR-5.4	The system shall offer a dark theme.	Could

3.6 CLI Access

ID	Requirement	Priority
FR-6.1	The CLI shall create a paste from a file argument or from standard input (for example, cat log.txt | ratcache) and print the paste's link.	Must
FR-6.2	The CLI shall accept flags for title, language, visibility, and expiration.	Must
FR-6.3	The CLI shall print a paste's raw content to standard output, given its ID or link.	Must
FR-6.4	The CLI shall let a user sign in, which stores an API token in a config file in the user's home directory.	Must
FR-6.5	The CLI shall list, edit, and delete a signed-in user's pastes.	Should
FR-6.6	The CLI shall exit with status 0 on success and a non-zero status on failure, so scripts can check the result.	Must

3.7 Administration

ID	Requirement	Priority
FR-7.1	The system shall let an administrator delete any paste.	Should
FR-7.2	The system shall let an administrator disable a user account.	Could

4. Non-Functional Requirements

4.1 Data Encryption

ID	Requirement	Priority
NFR-1.1	The system shall serve all traffic over HTTPS (TLS 1.2 or later) and redirect HTTP requests to HTTPS.	Must
NFR-1.2	The server shall encrypt the content and title of every paste, including anonymous pastes, with AES-256-GCM before storing them.	Must
NFR-1.3	The server shall use one server-wide encryption key for all pastes and store it outside the database and source code, so a stolen copy of the database cannot be decrypted.	Must
NFR-1.4	The system shall never write paste content or passwords to logs.	Must

4.2 Password Security

ID	Requirement	Priority
NFR-2.1	The system shall hash passwords with Argon2id or bcrypt (cost factor 12 or higher) and never store plaintext passwords.	Must
NFR-2.2	The system shall require passwords of at least 12 characters.	Must
NFR-2.3	The system shall lock sign-in for an account for 15 minutes after 5 failed attempts in a row.	Should
NFR-2.4	The system shall return the same error for a wrong username and a wrong password.	Must
NFR-2.5	The system shall store session cookies with the HttpOnly, Secure, and SameSite flags and protect forms against CSRF.	Must
NFR-2.6	The system shall store only hashes of API tokens, never the tokens themselves.	Should

4.3 Input Validation

ID	Requirement	Priority
NFR-3.1	The system shall reject paste content larger than 1 MB for anonymous users, or 5 MB for registered users.	Must
NFR-3.2	The system shall reject empty pastes.	Must
NFR-3.3	The system shall HTML-escape all paste content and titles before rendering them, to prevent cross-site scripting.	Must

NFR-3.4	The system shall use parameterized database queries for all user input, to prevent SQL injection.	Must
NFR-3.5	The system shall validate every field on the server (username format, email format, language and expiration values in their allowed lists), even when the client already checked it.	Must
NFR-3.6	The system shall rate-limit paste creation to 10 pastes per minute per IP address for anonymous users.	Should

4.4 Error Handling

ID	Requirement	Priority
NFR-4.1	The API shall return standard HTTP status codes (400, 401, 403, 404, 410, 413, 429, 500) with a JSON body holding an error code and a plain-language message.	Must
NFR-4.2	The system shall show the same "not found" page for a missing, expired, or private paste, so outsiders cannot tell which one it is.	Must
NFR-4.3	The system shall never show stack traces or internal details to users.	Must
NFR-4.4	The system shall log server errors with a timestamp and request ID.	Must
NFR-4.5	The CLI shall print readable error messages to standard error.	Must

4.5 Performance and Reliability

ID	Requirement	Priority
NFR-5.1	The server shall respond to 95% of paste create and view requests in under 500 ms with 50 concurrent users.	Should
NFR-5.2	Paste IDs shall be random and at least 8 characters long, so that links cannot be guessed.	Must
NFR-5.3	The database shall be backed up daily.	Could

4.6 Maintainability

ID	Requirement	Priority
NFR-6.1	The repo shall include setup instructions that let a new developer run the system locally in under 30 minutes.	Should
NFR-6.2	Automated tests shall cover every API endpoint.	Should

Project links:

Github: https://github.com/mkmcgrath/ratnest

Jira Board (requires login)
https://ratnest.atlassian.net/jira/software/projects/KAN/boards/2